



Faculty of Science and Technology

Programme Structure & Syllabus

For

First Year

Master of Science

Computer Science

(MSC CS)

P-2024

With Effect from Academic Year 2025– 2026

	Authority	Date
Proposed by	Board of Studies in Computer Science	
Approved by	Academic Council, Vishwakarma University, Pune	

Issued by

Chairman – BoS

Dean of Faculty

Director, IQAC

MSCCS24101: Design and Analysis of Algorithm

Course Type	DSC	Semester	I
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3 Hr./Week	Lecture	3	CIE Marks	60
Tutorial	1 Hr./Week	Tutorial	1	ESE Marks	40
Practical/Studio	... Hr./Week	Practical/Studio		Total Marks	100
Total	... Hr./Week	Total			

Course Description

The overall objective of this course is to cater the need of computational field. The content of this course are according to the current trends of research in computer science and requirements of industry expectations. Some courses of this program are exclusively designed towards development of analytical, presentation and personality development skills among the students, through which the students get prepared and trend for building their career in computer science and it related to applied technology, research and development.

Course Outcomes

CO No.	Statement
1	Analyze worst-case running times of algorithms using asymptotic analysis.
2	Compare between different data structures. Pick an appropriate data structure for a design situation.
3	Ability to design algorithms using standard paradigms like: Greedy, Divide and Conquer, Dynamic Programming and Backtracking.
4	Able to Explain the major graph algorithms and Employ graphs to model engineering problems, when appropriate.
5	Able to Compare between different data structures and pick an appropriate data structure for a design situation.

Mapping of COs to POs and PSOs

CO Number	POs										PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PSO 1	PSO 2	PSO 3	PSO 4	BTL
CO1	3	1	2								3	2	1	1	1,2
CO2		2	3	1					2	1	2	1	2	3	2,4
CO3				3	2	1		1		3	2	1	4	4	1,3,4
CO4			2	3	1				3		1	3	3	2	5,6
..															

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No I	Introduction to Algorithms	10 Hours	CO	BTL
Algorithm definition and characteristics, Space complexity, Time complexity- worst case, best case, average case, Complexity, asymptotic notation, Recursive and non-recursive algorithms, Sorting algorithms : insertion sort, heap sort, bubble sort, Sorting in linear time: counting sort, concept of bucket and radix sort Searching algorithms: Linear, Binary			1	1,3
Unit No II	Divide and Conquer strategy and Greedy Methods:	14 Hours	2	1,2,4
General method, control abstraction, Binary search, Merge sort, Quick sort, Comparison between Traditional Method of Matrix Multiplication vs. Strassen's Matrix Multiplication, Writing simple algorithm using Divide and conquer strategy: power(x,n), find occurrence of a number from array of N integers, to find minimum from an array, mini-max algorithm, largest number multiplication, simple convex algorithm Greedy Method: Knapsack problem, Job sequencing with deadlines, Minimum-cost spanning trees: Kruskal and Prim's algorithm, Optimal storage on tapes, Optimal merge patterns, Huffman coding, Shortest Path :Dijkstra's Algorithm				
Unit No III	Dynamic Programming and Decrease and conquer:	16 Hours	3	3,4,5
Principle of optimality, Matrix chain multiplication, 0/1 Knapsack Problem i)Merge & Purge ii)Functional Method, Bellman Ford Algorithm, All pairs Shortest Path Floyd-Warshall Algorithm, Travelling Salesperson problem, Longest common subsequence, String editing Decrease and conquer: Definition of Graph Representation of Graph, By Constant - DFS and BFS, Topological sorting, Strongly Connected components and spanning trees, Articulation Point and Bridge edge				
Unit No IV	Backtracking, Branch and Bound Techniques	16 Hours	4	3,4,5
General method, Fixed Tuple vs. Variable Tuple Formulation, n- Queen's problem, Graph colouring problem, Hamiltonian cycle, Sum of subsets Branch and Bound Techniques: Introduction : Branch and bound terms like definition of live node, E-node, Dead node, Least cost (LC) search, Least cost Branch and Bound (LCBB) 0/1 knapsack problem using LCBB method (fixed tuple size), Travelling Salesman problem using LCBB method (variable tuple size)				
Unit No V	Problem Classification	4 Hours	5	2,3
The class of P, NP, NP-hard and NP –Complete, Relationship among P class, NP class, NP-hard and NP –Complete, Cook's theorem				

Textbooks

1	<i>Introduction to Algorithms, Thomas H.Cormen, Charles E.Leiserson, Ronald L. Rivest and Clifford Stein, Third Edition, PHI Learning Private Limited, 2012.</i>
2	<i>Data Structures and Algorithms, Alfred V. Aho, John E. Hopcroft and Jeffrey D. Ullman, Pearson Education, Reprint 2006.</i>
3	<i>Algorithms Design and Analysis, Harsh Bhasin, Oxford university press, 2016.</i>
4	<i>Design and Analysis of Algorithms, S. Sridhar, Oxford university press, 2014.</i>